

Generative Adversarial Networks (GAN)

One way to teach a system about the character of a data set is to pair it with another system that's trying to trick it.

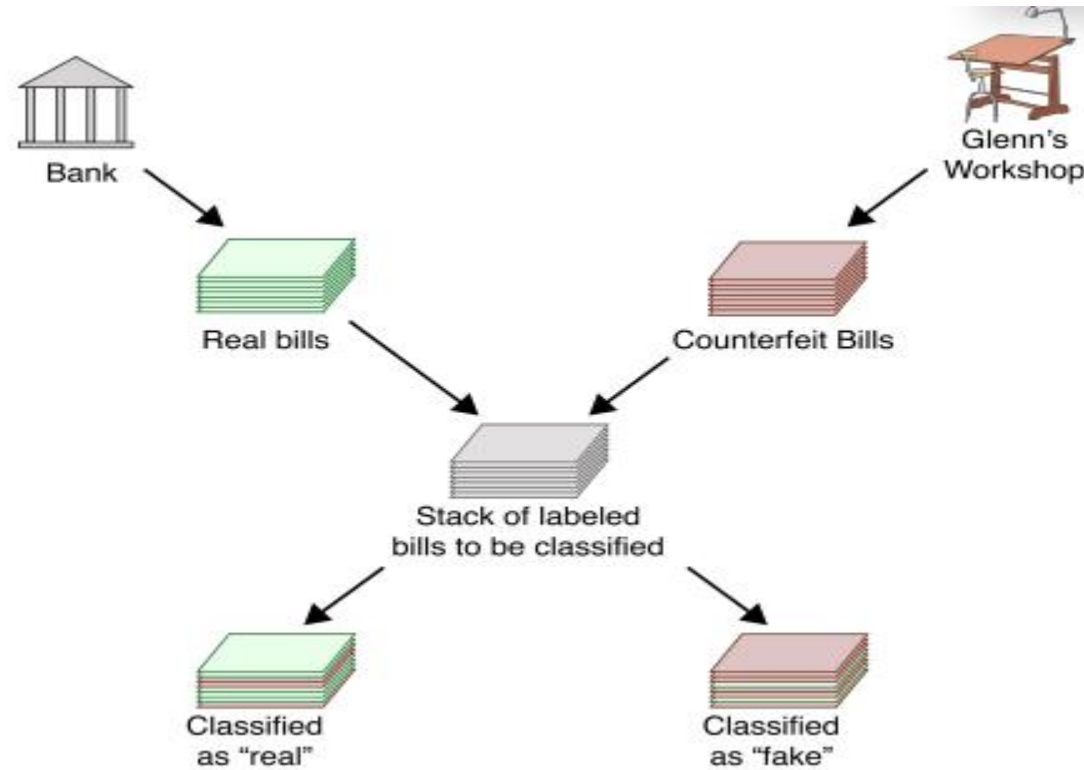
When our learner can distinguish real data from forgeries,

we can use it to make more data like the input.

A Metaphor: Forging Money

- ❑ The story begins with two conspirators, whom we'll call Glenn and Dawn.
- ❑ Glenn's name starts with G because he'll be playing the role of the generator, in this case forging new money.
- ❑ Dawn's name begins with D because she'll be our discriminator, tasked with determining whether any given bill is real, or one of Glenn's forgeries.
- ❑ They're going to work together so that they both become as good as possible at their jobs, in the process forcing each other to become better and better.

A Metaphor: Forging Money



Methodology

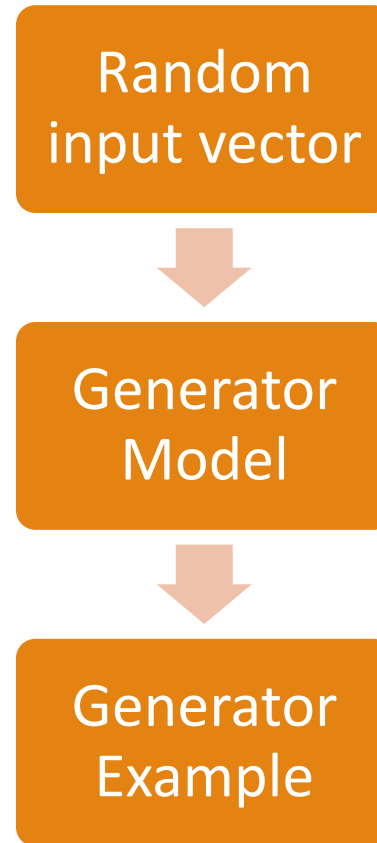
Generator

- give some training sample called REAL sample
- Based on the real sample generator will try to create a fake sample
- fed it to the discriminator
- and will try to fool it until the discriminator returns it as real sample.

Discriminator

- try to find whether the sample given by the generator is real or fake.
- If it thoughts fake sample error then again generator will try to modify the fake sample
- and fed it into the discriminator.
- The cycle will continue until it thoughts real sample error.

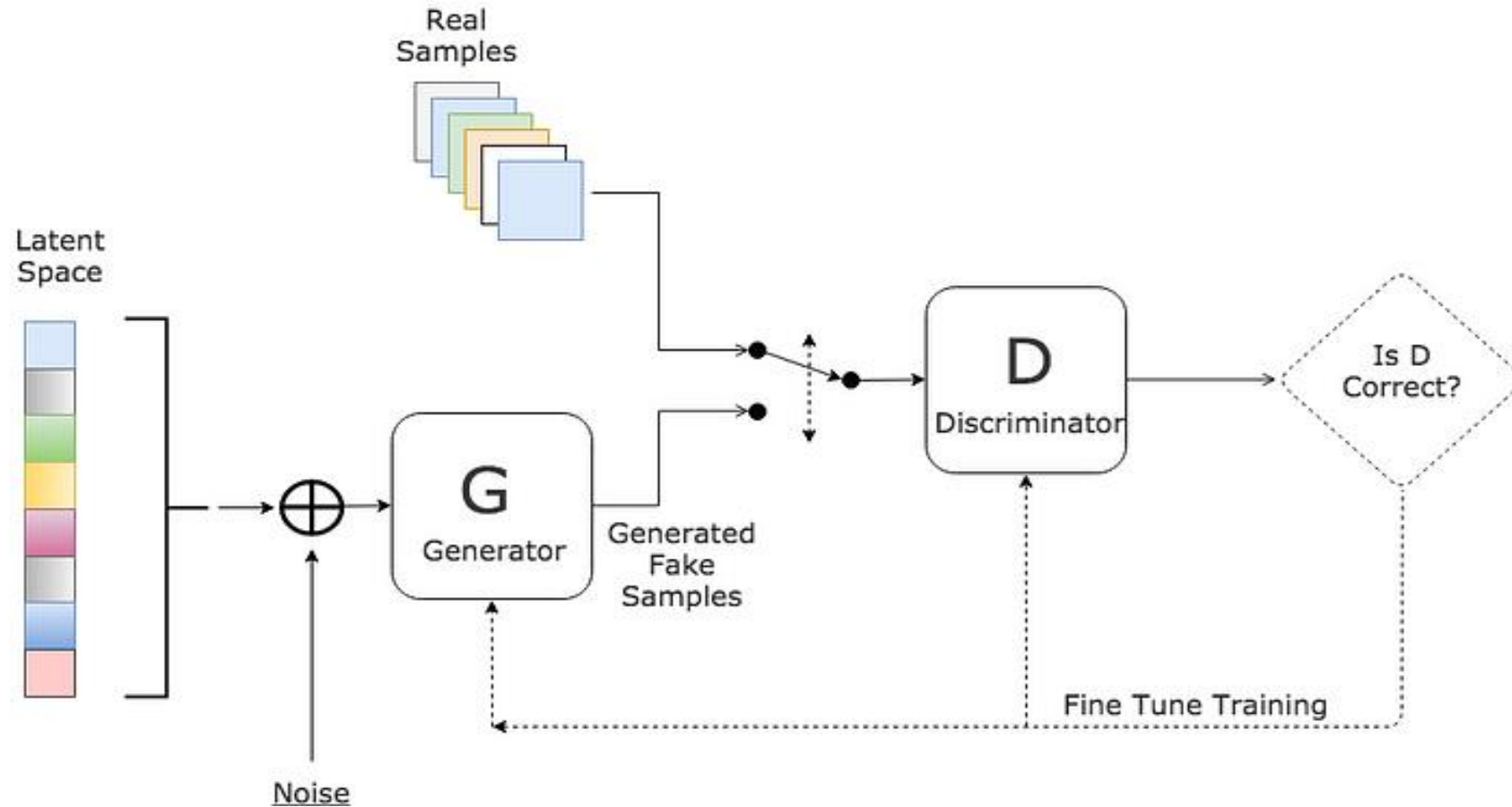
The Generator Model



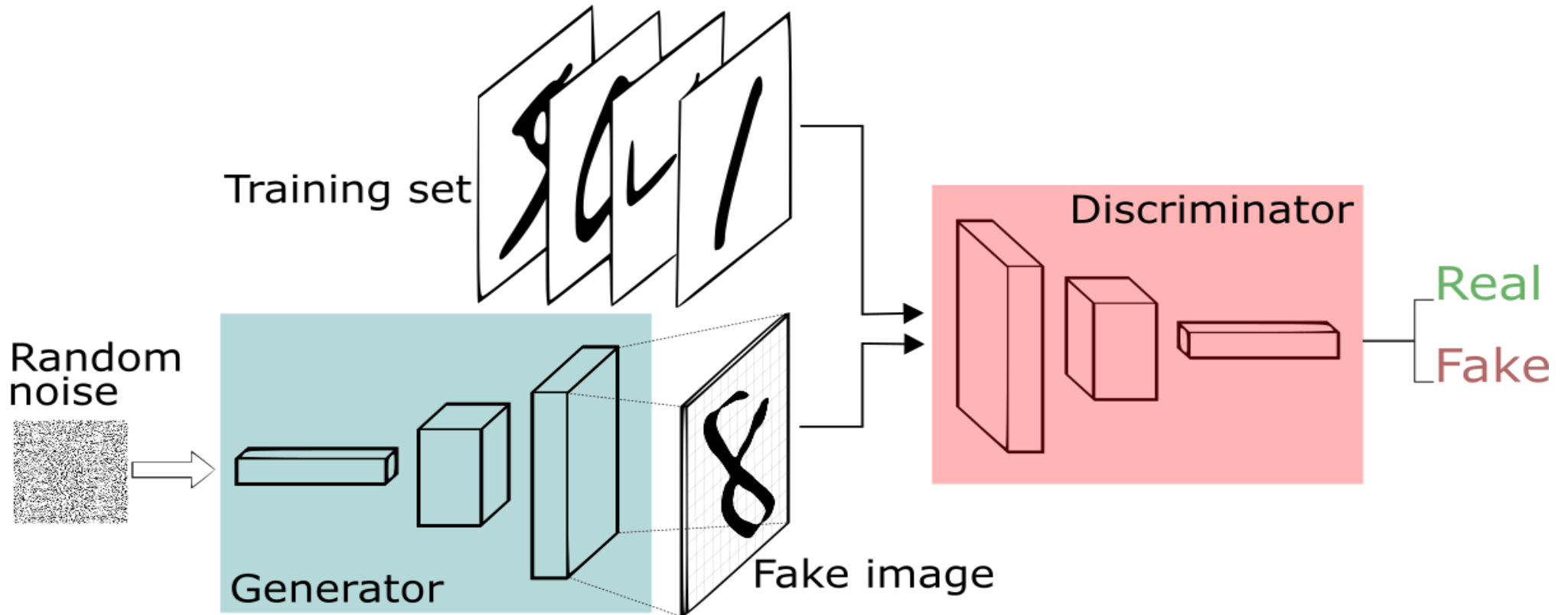
The Discriminator Model



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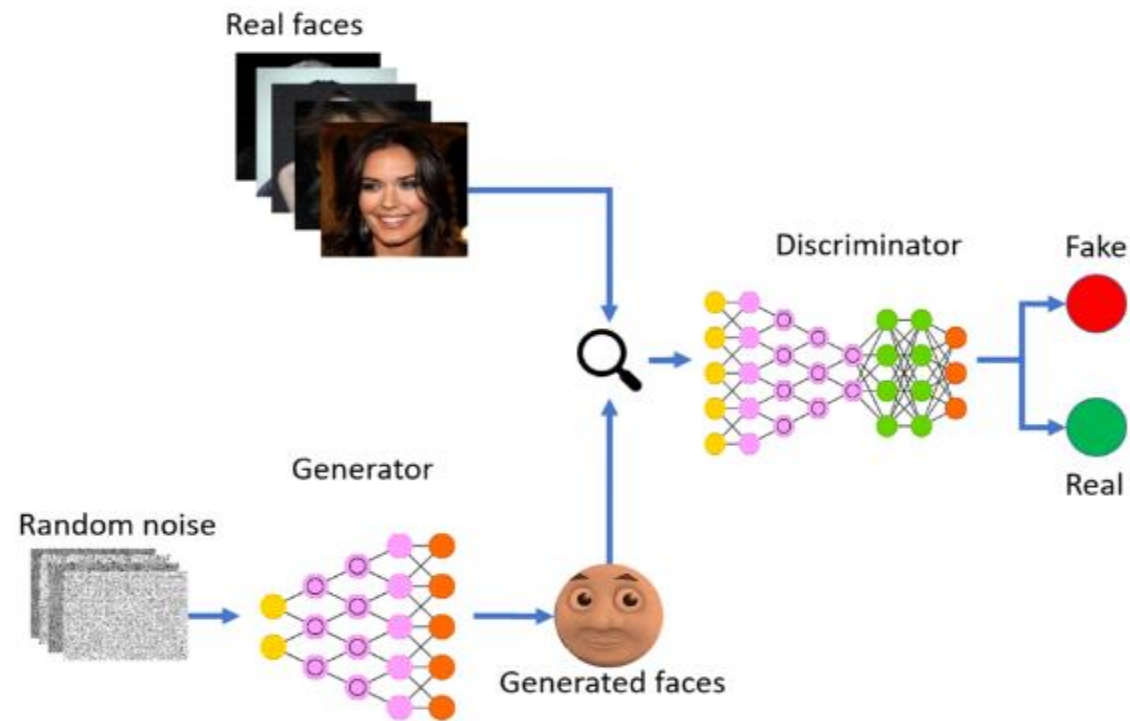
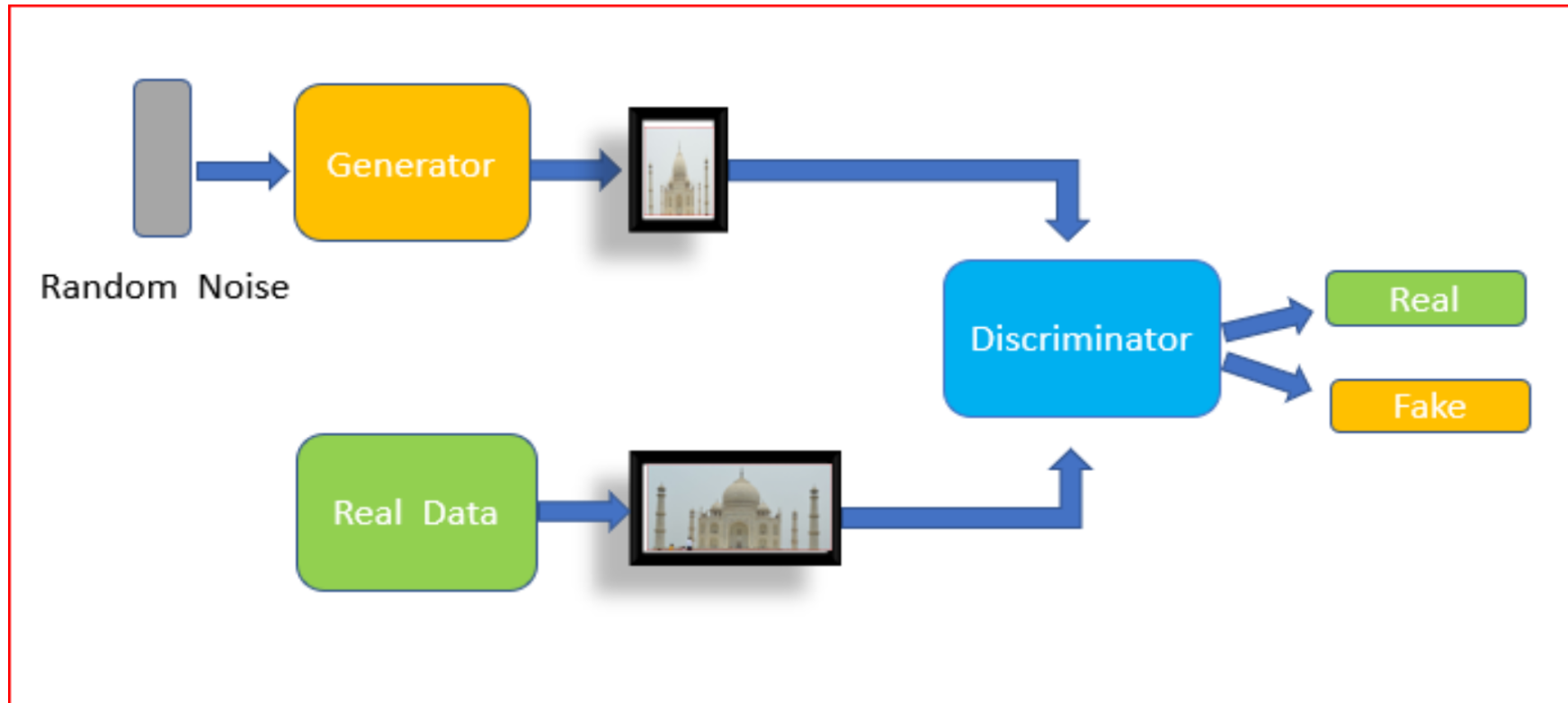


Image Generation Using GAN

- The goal of the generator is to *generate* or produce new and synthetic samples of a certain input, which could be a random set of values or noise.
- For instance, if you trained it with the class of a cat, the generator would perform a series of computations, and then produce an image of a cat, which isn't real but looks real

Image Generation Using GAN



DeepFakeUsing GAN

- By using neural networks, GANs can make a significant impact on any industry dealing with data and images. This technology showed that there is a possibility to generate realistic fake photos or replace people's faces with other ones.
- This phenomenon was later dubbed deepfake (deep learning + fake).
- **GAN is the popular method that is being used to create deepfakes from a given dataset.**

DeepFakeUsing GAN

